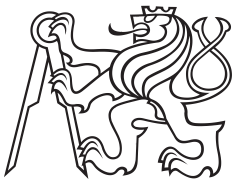

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Visual Object Recognition with Unknown Input Detection

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Abstract

Keywords: Out-of-Distribution Detection (OOD), Visual Object Recognition, Vision Transformers (ViTs), DINO, Post-Hoc Methods, Hierarchical Detection, Feature Representation, Unknown Input Detection

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Abstrakt

Klíčová slova: Detekci mimo distribuci (OOD), Rozpoznávání vizuálních objektů, Vizuální transformátory (ViTs), DINO, Post-hoc metody, Hierarchická detekce, Reprezentace příznaků, Detekce neznámých vstupů

Překlad názvu: Rozpoznávání vizuálních objektů s detekcí neznámých vstupů

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Chapter 1

Introduction

Deep learning architectures are currently experiencing a substantial surge in prevalence. Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) have demonstrated remarkable performance in various computer vision tasks. However, the real world is full of unknowns, which poses serious challenges for these models. When deployed in dynamic environments, they often encounter inputs that differ significantly from their training data, leading to unreliable predictions, **as illustrated in Figure 1.1.** The task of identifying such inputs, known as out-of-distribution (OOD) detection, is crucial for ensuring the safety and reliability of AI systems.

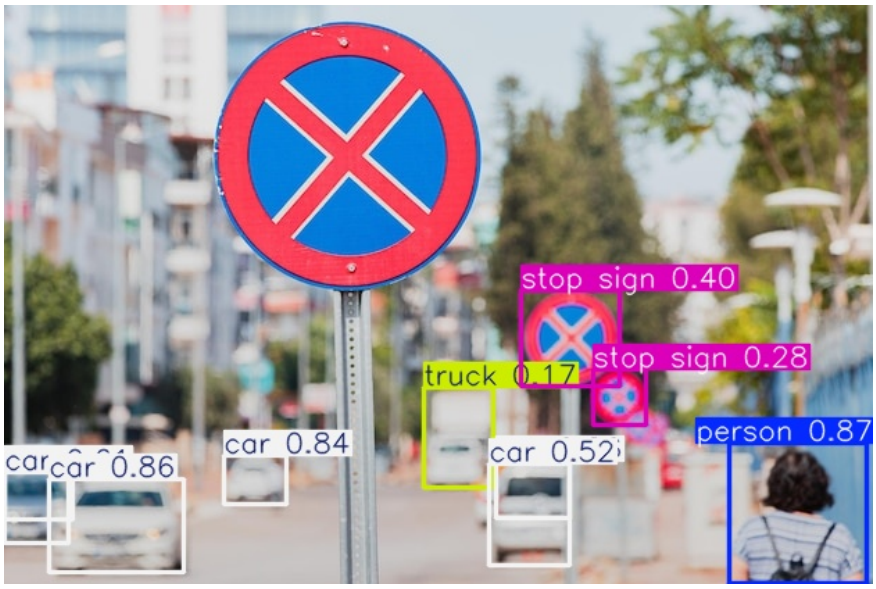


Figure 1.1: Limitations of a standard YOLO detector. The model fails to recognize the nearest 'No Parking' sign (False Negative) while misclassifying distant ones as 'Stop Signs'. This illustrates the necessity for robust feature representations.

Previous research has primarily focused on binary OOD detection, where models must distinguish only between in-distribution (ID) and OOD samples.[1] While this is an important step towards robust AI, it does not fully capture the complexity of real-world scenarios. For example, when an au-

tonomous vehicle encounters an unseen input, it is insufficient to simply flag it as OOD. Instead, a hierarchical approach assigns it to a generic class (e.g., 'obstacle'), preserving semantic information essential for decision-making.

To address these challenges, this thesis explores the capabilities of self-supervised Vision Transformers, specifically the DINOv3 architecture[2]. We selected this model for its state-of-the-art performance on out-of-distribution benchmarks (as shown in ObjectNet evaluations) and its ability to produce rich semantic representations while being kept frozen. This "frozen" capability allows for substantial computational savings, making the model ideal for efficient **post-hoc** analysis without the need for extensive fine-tuning.

1.1 Goals

The primary objective of this thesis is to analyze DINOv3 internal representations and evaluate their suitability for hierarchical classification. The specific goals are:

1. **Feature Space Visualization:** To implement a feature extraction pipeline and analyze class separability using PCA and t-SNE dimensionality reduction on datasets such as CIFAR and Oxford-IIIT Pet.
2. **Attention Mechanism Analysis:** To visualize self-attention maps and interpret how the model spatially isolates semantic objects from the background without supervision.
3. **Layer-wise Evolution:** To track the trajectory of [CLS] tokens across network layers, identifying the specific depth where the model begins to separate distinct semantic concepts.

Chapter 2

Related Work

This chapter reviews the existing literature relevant to the thesis. It covers the evolution of visual backbone architectures focused on Vision Transformers and Self-Supervised Learning. In due course, it delves into established methodologies for OOD detection. Finally, it explores the emerging concept of hierarchical OOD detection, highlighting its significance and the challenges it presents.

2.1 Visual Representation Learning

State-of-the-art computer vision architectures have evolved significantly from traditional CNNs to more sophisticated models like ViTs.

2.1.1 Vision Transformers (ViT)

Vision Transformers (ViTs)[3] have revolutionized and dethroned CNNs as the progressively increasing architecture in computer vision. Unlike CNNs, which process images via local receptive fields and convolutional layers, ViTs leverage the self-attention mechanism from the Transformer architecture[4] to capture global dependencies in images.

2.1.2 Self-Supervised Learning and DINO

Training ViTs typically requires massive labeled datasets. To mitigate this, Self-Supervised Learning (SSL) has emerged as a powerful alternative. Methods like DINO (Distillation with No Labels)[5] and its successors, DINOv2 and DINOv3[2][6], learn robust features by enforcing consistency between different views of the same object.

2.2 Out-of-Distribution Detection

The aim of OOD detection is to safely distinguish between data that the model has seen during training (ID) and data that is novel or anomalous (OOD).

2.2.1 Post-Hoc OOD Methods

Post-hoc OOD detection methods derive anomaly scores directly from a pre-trained model without the need for parameter modification or retraining. This property makes them computationally efficient and easily deployable in real-world scenarios.

Maximum Softmax Probability (MSP): The fundamental approach proposed by Hendrycks and Gimpel[7] assumes that a classifier's softmax output can be used to gauge uncertainty. Inputs with low maximum softmax probabilities are flagged as OOD, indicating that the model is less confident in its predictions.

Distance based Methods: To mitigate overconfidence, these methods assess the distance of an input's representation from known class distributions. Lee at al.[8] proposed using the **Mahalanobis Distance**, modeling the pre-softmax features of each class as a multivariate Gaussian distribution. An input is flagged as OOD if it lies far from the nearest class centroid in this latent space.

Energy-Based Models: Liu et al.[9] introduced an energy-based scoring function derived from the model's logits. The energy score provides a more calibrated measure of uncertainty, with lower energy values indicating higher confidence in the input being ID.

ODIN (Out-of-Distribution detector for Neural networks): Liang et al.[10] introduced ODIN, which improves upon the MSP baseline by combining two strategies: *temperature scaling* and *input perturbation*.

- First, a large temperature scalar is applied to the softmax function to separate the probability distributions of ID and OOD samples.
- Second, small perturbations are added to the input image in the direction of the gradient to aggressively maximize the predicted softmax score for ID samples.

While effective, the requirement for input preprocessing increases computational cost during inference compared to pure output-based methods.

2.2.2 Training Based OOD Methods

While post-hoc methods operate on fixed models, training-based methods explicitly optimize the model parameters to enhance the separability between ID and OOD data during the learning phase.

Ensemble Methods: Deep Ensembles[11] utilizes multiple models trained independently to capture uncertainty. The variance in predictions across the ensemble serves as an indicator of OOD inputs, with higher variance suggesting greater uncertainty. While this approach significantly improves uncertainty estimation and OOD detection accuracy, it comes with a high computational cost (both for training and inference) that is often prohibitive for real-time applications.

Regularized Training: Other methods introduce specific loss functions or *proxy* OOD data during training. For instance, CSI [12] utilizes contrastive

learning with strong data augmentations to simulate distributional shifts. Similarly, Adversarial Reciprocal Points Learning (ARPL) [13] generates *reciprocal points* in the latent space to represent the unknown class. Although ARPL achieves state-of-the-art performance on many benchmarks, it requires a complex training scheme and extensive hyper-parameter tuning [14].

Outlier Exposure (OE): A direct approach involves training the model on both ID data and a large auxiliary dataset of known outliers (e.g., 80 Million Tiny Images). The model is penalized if it predicts high confidence for these outliers [15]. While effective, this method relies on the availability of a curated OOD dataset and may not generalize well to anomalies that are distributionally distinct from the auxiliary data.

2.3 Hierarchical and Open-Set Recognition

Standard OOD treats all unknown inputs as a single homogeneous class. Nevertheless, in many real-world applications it is beneficial to understand semantic nature of unrecognized objects.

2.3.1 From Open-Set to Open-World

Open-Set Recognition (OSR) extends long-established classification by requiring models to accurately classify known classes while rejecting unknown classes during inference[16].

Open-World Recognition (OWR) inclines to more a realistic approach, where the system must identify novel inputs and eventually incorporate them into its knowledge base through continuous learning[17].

2.3.2 Hierarchical Recognition

In many real-world applications, visual concepts naturally exhibit a taxonomic organization. For instance, in biological classification, individual species are grouped into genera, families, and orders. Hierarchical recognition leverages this inherent structure to provide predictions that are semantically meaningful even when fine-grained identification fails.

Standard classification models typically employ a flat label space, treating all misclassifications with equal severity. In this setting, confusing a *dog* with a *wolf* incurs the same penalty as confusing a *dog* with a *car*, despite the vast semantic discrepancy between these errors. This limitation was notably addressed by Bertinetto et al. [18], who argued that models should aim for "better mistakes" by minimizing hierarchical distance rather than just 0/1 loss.

Hierarchical models utilize structured label spaces, often derived from ontologies such as WordNet [19], to enable *coarse-to-fine* inference. As illustrated in Figure 2.1 using the Oxford-IIIT Pet dataset [20], distinct categories (e.g., specific breeds) can be naturally aggregated into super-categories (e.g., *Felines* vs. *Canines*). In the context of OOD detection, this

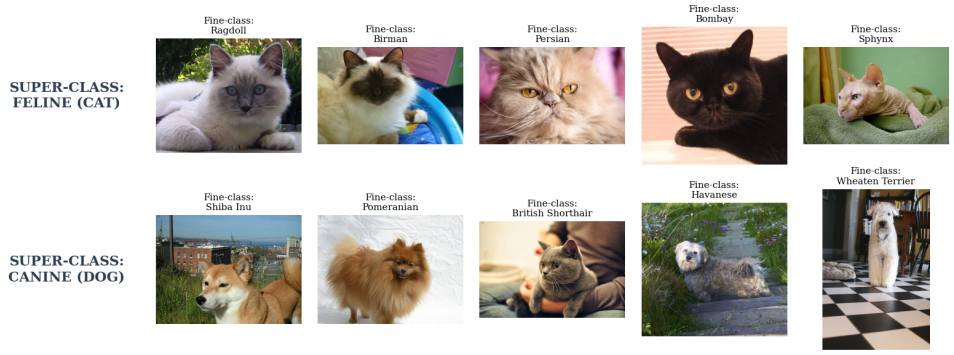


Figure 2.1: Visual demonstration of semantic hierarchy. Fine-grained categories (specific breeds) naturally cluster into broader super-classes (Felines and Canines). A hierarchical OOD detector can leverage this structure to classify an unknown dog breed as a generic 'Canine' rather than rejecting it completely.

structure provides a robust fallback mechanism. If the model encounters a novel input that implies high uncertainty at the leaf level, it can back-off to a parent node (e.g., classifying an unknown animal simply as *Animal*), thereby preserving system reliability and safety.

Chapter 3

Methodology and Analysis

The initial phase of this research consisted of assessing the efficacy of DINOv3 feature representations for our OOD problem. A critical requirement for our task is the model's ability to enforce tight grouping of semantically similar classes in the feature space while maintaining separability from distinct concepts.

To verify this property, we extracted feature embeddings from diverse datasets and applied dimensionality reduction techniques such as Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE).

3.1 Dataset Selection

To evaluate the robustness and generalizability of the features learned by the DINOv3 model, we employed a suite of standard computer vision benchmark datasets, alongside a custom dataset to test domain shifts:

- **CIFAR-10** [21]: A widely utilized benchmark consisting of 60,000 32×32 color images across 10 distinct classes, with 6,000 images per class. The standard split includes 50,000 training images and 10,000 test images.
- **CIFAR-100** [21]: Similar in structure to CIFAR-10 but significantly more challenging due to its fine-grained nature. It contains 100 classes with 600 images each, split into 500 training and 100 testing images per class.
- **SVHN (Street View House Numbers)** [22]: A real-world image dataset designed for developing machine learning and object recognition algorithms. It comprises over 600,000 labeled digits $(0 - 9)$ cropped from house numbers in Google Street View imagery, presenting a distinct distribution shift from standard object datasets.
- **Food-101** [23]: A large-scale dataset for fine-grained food recognition, consisting of 101,000 images representing 101 different food categories. Each category contains 1,000 images, offering a high variance in visual appearance within classes.

- **Oxford-IIIT Pet** [20]: A dataset containing 7,349 images of 37 different breeds of cats and dogs. Each image is annotated with a breed label and a pixel-level segmentation mask, making it suitable for both classification and segmentation tasks.
- **Pokémon Dataset**: A custom dataset containing images of various Pokémon species. This dataset was introduced to evaluate the model's ability to generalize to unseen, synthetic classes that differ significantly from natural images.

3.2 Feature representation via PCA

Prior to investigating the layer-wise evolution of features within the DINOv3 architecture, we conducted an initial assessment of the overall quality of the learned representations. This involved extracting feature embeddings from the final layer of the model for images across the selected datasets.

3.2.1 Dimensionality Reduction and Visualization

To visuallize the high-dimensional feature space of the data and demonstrate class separability, we employed **Principal Component Analysis (PCA)** to reduce the dimensionality of the embeddings from $\mathbb{R}^{1024}$ to $\mathbb{R}^2$ and $\mathbb{R}^3$.

The goal was to project the *final* representations of either the entire dataset or a subset onto a common manifold. This visualization helped us to confirm the following properties:

- **Intra-class Compactness:** Samples of the same class are grouped tightly together.
- **Inter-class Separability:** Different classes form distinct, non-overlapping clusters.

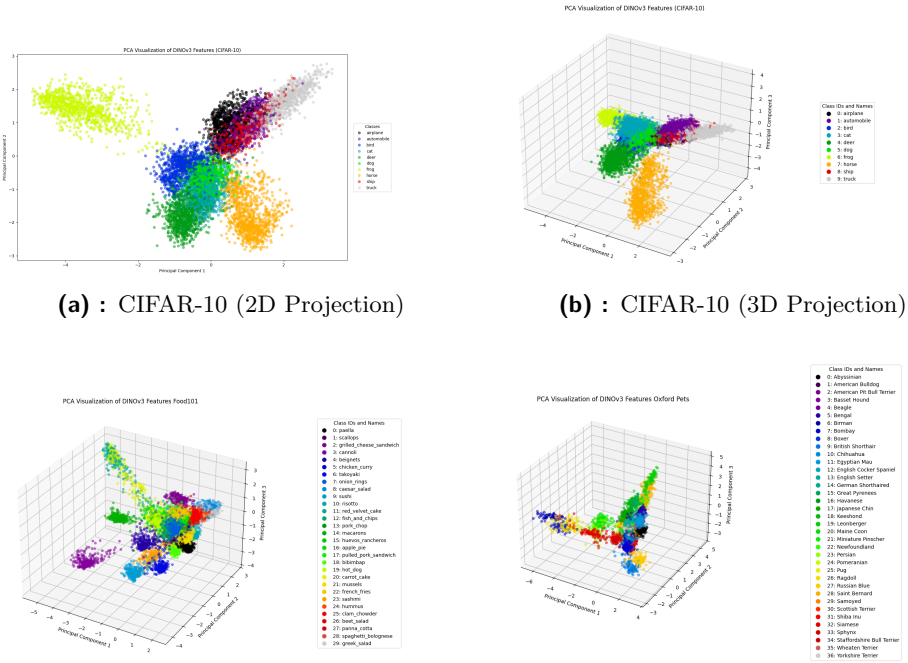
3.3 Visual Analysis of Feature Distributions via PCA

3.3.1 General Clustering Performance

For the majority of the evaluated datasets such as CIFAR-10, FOOD101 and Oxford-IIIT Pet, the visualization reveals a strong class-separation capability as illustrated in Figure 3.1. The class centroids form distinct clusters with minimal overlap, indicating that the DINOv3 model has learned robust and discriminative features.

3.3.2 Challenges in Cross-Domain Representation

The analysis of the Pokémon dataset presents a more complex distribution, as illustrated in Figure 3.2. While some classes form distinct clusters, the



(a) : CIFAR-10 (2D Projection)

(b) : CIFAR-10 (3D Projection)

(c) : Food101 (3D Projection)

(d) : Oxford-IIIT Pet (3D Projection)

Figure 3.1: PCA visualizations of feature space separation across standard benchmarks. **(a)** and **(b)** compare 2D vs. 3D projections for CIFAR-10, demonstrating clear intra-class cohesion. **(c)** and **(d)** extend this analysis to Food101 and Oxford-IIIT Pet, confirming the model's ability to maintain distinct semantic clusters across diverse domains.

boundaries are significantly not as well-defined as in the natural image datasets.

We attribute this behavior to high **intra-class domain variance**. Unlike standard datasets where each class has a relatively consistent visual appearance, the Pokémon dataset mixes heterogeneous sources:

- **Stylistic variance:** 2D sprites, 3D renders, anime frames, and real-world photos of merchandise.
- **Feature mismatch:** Visually dissimilar instances can share the same label, reducing embedding compactness.

Consequently, the model likely creates multiple sub-clusters for a single class based on these artistic domains rather than merging them into a single unified entity, resulting in the observed "sketchy" separation.

3.3.3 Failure Case: Street View House Numbers (SVHN)

A notable deviation is observed with the SVHN dataset, where PCA fails to separate digit classes $(0-9)$, forming a single overlapped cloud as shown in Figure 3.3.

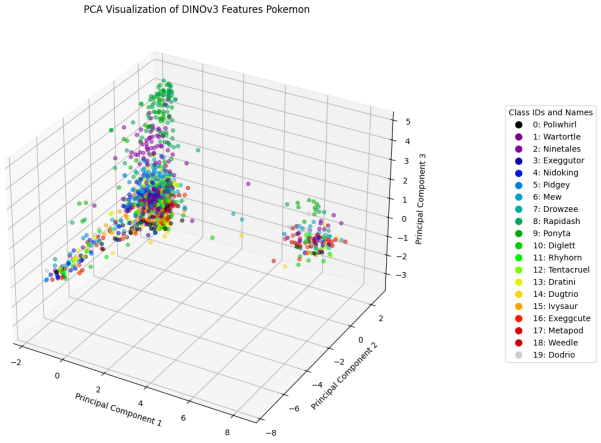
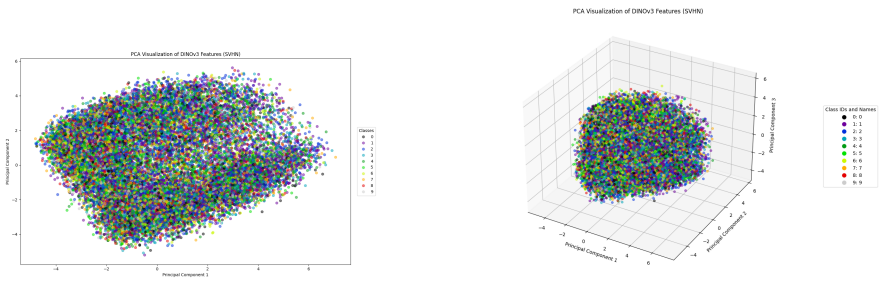


Figure 3.2: Feature space distribution of the Pokémon dataset. The clusters exhibit higher variance and less distinct separation compared to natural images. This is attributed to the dataset's multi-domain nature, containing stylistically divergent samples (e.g., pixel art vs. plush toys) within the same semantic category.

We attribute this failure to the model's inherent **texture bias**. Unlike natural objects, digits are defined purely by geometry rather than semantic textures (e.g., fur or foliage). Furthermore, background clutter often occupies more area than the digit itself. Without fine-tuning, the encoder likely prioritizes these background textures over the structural signal of the class label.



(a) : SVHN (2D Projection) **(b) :** SVHN (3D Projection)

Figure 3.3: Feature space visualization of the SVHN dataset. Both 2D and 3D projections show significant class overlap, indicating the model's inability to linearly separate geometric digits from background texture without specific fine-tuning.

3.4 Feature representation via t-SNE

While PCA effectively demonstrates the separability of classes, it is inherently a linear technique. To ensure that we did not overlook complex, non-linear structures in the feature space, we also applied **t-Distributed Stochastic**

Neighbor Embedding (t-SNE) [24].

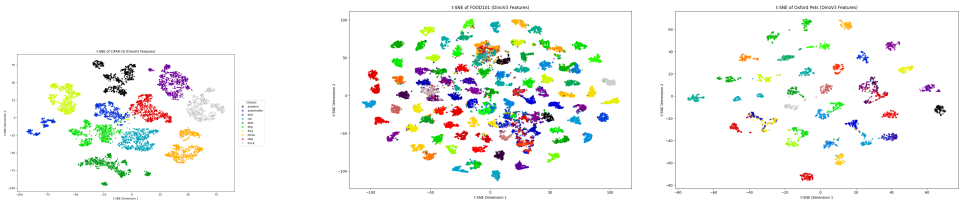
3.4.1 Motivation and Methodology

Unlike PCA, t-SNE is a probabilistic technique designed to preserve local neighborhood structures. It is particularly effective at revealing clusters in data where the relationship between classes is non-linear. We hypothesized that t-SNE might uncover finer sub-structures or better resolve classes that appeared only weakly separated in the linear PCA projection (e.g., some Pokémon classes).

3.4.2 Comparative Analysis: Cleaner Separation and Non-Linearity

The application of t-SNE yielded a distinct visual improvement over PCA, producing significantly cleaner and more compact clusters. By prioritizing local neighborhood preservation, t-SNE was able to resolve boundaries that appeared diffuse in the linear projection.

General Benchmarks: For standard datasets like CIFAR-10, Food-101, and Oxford-IIIT Pet, the clusters became tighter and more isolated, reinforcing the robust semantic separability of the DINOv3 features. This general performance is visualized in Figure 3.4.

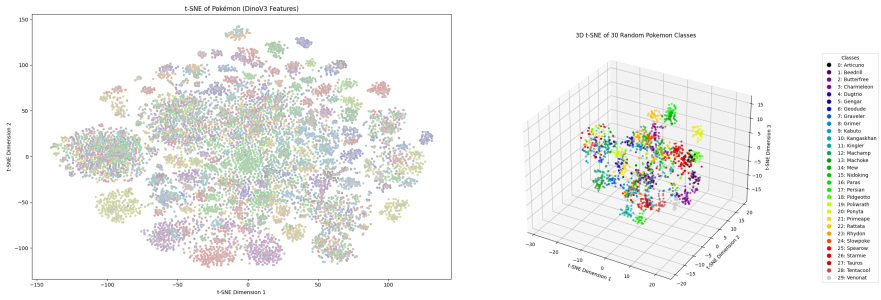


(a) : CIFAR-10 **(b) :** Food-101 **(c) :** Oxford Pets

Figure 3.4: t-SNE visualizations for CIFAR-10, Food-101, and Oxford-IIIT Pet. Compared to PCA, the non-linear embedding produces significantly tighter clusters with cleaner boundaries, demonstrating the high quality of the feature representations.

The Pokémon Case: The most notable improvement was observed in the Pokémon dataset (Figure 3.5). While PCA struggled with the high domain variance (mixing sprites, 3D renders, and plushies), t-SNE successfully disentangled these non-linear relationships. The resulting visualization shows clear, distinct clusters for different Pokémon species, suggesting that while the domain shift introduces non-linearity, the semantic identity remains locally consistent.

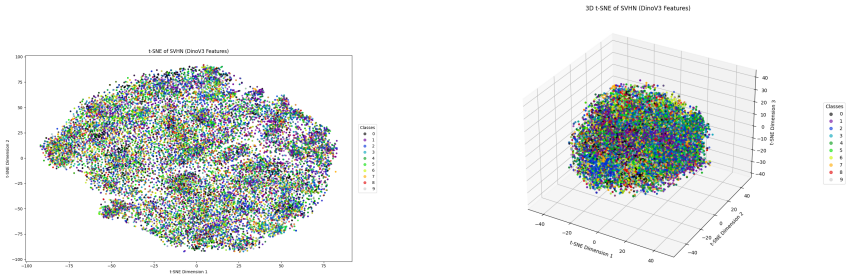
Persistent Failure on SVHN: Conversely, the SVHN dataset (Figure 3.6) remained inseparable. Even under non-linear embedding, the digit classes overlap in a single amorphous cloud. The persistence of this failure across both linear (PCA) and non-linear (t-SNE) techniques confirms that the issue is not a projection artifact but a fundamental lack of discriminatory signal



(a) : Pokémon (2D t-SNE) **(b) :** Pokémon (3D t-SNE)

Figure 3.5: t-SNE visualization of the Pokémon dataset. Unlike the linear PCA projection, t-SNE successfully resolves the non-linear domain shifts, separating the classes into distinct islands.

in the features, driven by the model's bias towards background texture over geometric shape.



(a) : SVHN (2D t-SNE) **(b) :** SVHN (3D t-SNE)

Figure 3.6: t-SNE visualization of the SVHN dataset. The failure to separate classes persists even with non-linear dimensionality reduction, confirming the lack of semantic separability in the underlying features.

3.5 Layer-wise Feature Evolution and Trajectory Analysis

To qualitatively analyze the internal dynamics of the DINOv3 architecture [2], we developed a visualization pipeline capable of mapping the high-dimensional latent space of 1024 dimensions into a 3-dimensional RGB color space. This process, which we refer to as a "Layer-wise Feature Trajectory," allows for the observation of semantic evolution as information propagates from the input layers to the deep semantic layers. The use of PCA to inspect the semantic layout of self-supervised features follows methodologies established in DINO-based research [2].

3.5.1 Data Extraction and Pre-processing

The extraction mechanism relies on registering *forward hooks* within the PyTorch framework. We iterate through the model's architecture to identify the transformer blocks (specifically `model.encoder.layer`) and intercept the output tensor $O^{(l)}$ at each layer $l \in \{0, \dots, L-1\}$.

For a given input image, the output at layer l is a tensor of shape $(1, N_{seq}, D)$, where N_{seq} represents the sequence length and D is the embedding dimension. To visualize the spatial layout of features, we discard the special tokens—specifically the `[CLS]` token and the register tokens retaining only the patch tokens. These patch tokens are then reshaped back into a 2D grid corresponding to the original image topology ($H \times W$).

3.5.2 Dimensionality Reduction Strategies

Visualizing $D = 1024$ dimensions requires projecting the data onto a lower-dimensional manifold. We implemented and compared two distinct strategies for this projection using Principal Component Analysis (PCA): *Global PCA* and *Local PCA*.

Method A: Global PCA (Trajectory Preservation)

In the Global approach, we prioritize the temporal and geometric consistency of the features across the network depth. We aggregate the feature vectors from all layers l and all spatial positions into a single global dataset $\mathbf{X}_{global}$:

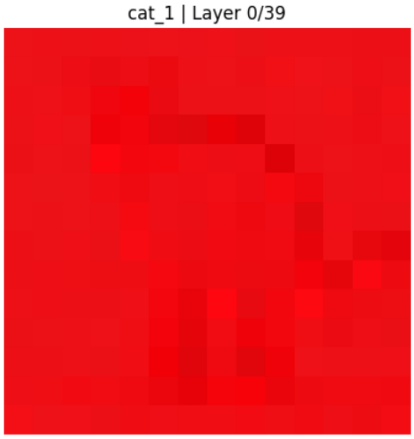
$$\mathbf{X}_{global} = \bigcup_{l=0}^{L-1} \mathbf{X}^{(l)} \tag{3.1}$$

To ensure that feature representations from different depths of the network are geometrically comparable, we compute a single, unified transformation matrix W_{global}. This is achieved by fitting the PCA estimator on the aggregated dataset $\mathbf{X}_{global}$, effectively forcing the algorithm to learn a "global dictionary" of variance that encompasses the model's entire inference process—from raw pixel statistics in shallow layers to high-level semantic abstractions in deep layers.

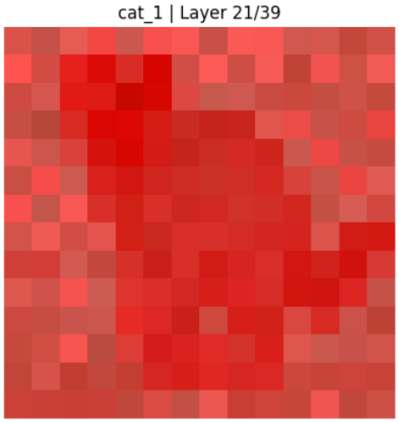
Mathematically, this identifies the three principal components that explain the maximum variance across the *full hierarchy* of the network. Once this fixed basis W_{global} is established, we project the specific feature tensor $\mathbf{X}^{(l)}$ of every individual layer l onto these static axes. By using a constant projection matrix for all layers, we ensure that the resulting color shifts in the visualization represent genuine geometric transformations of the feature vectors (e.g., the data manifold rotating from a "pixel-correlation" subspace to a "semantic" subspace) rather than arbitrary statistical realignments between frames.

Visual Outcome: This method results in a smooth color progression. We observed a distinct shift from red hues in initial layers to azure/cyan hues in deeper layers. This indicates that the feature subspace occupied by raw visual patterns (shallow layers) is mathematically orthogonal to the subspace of

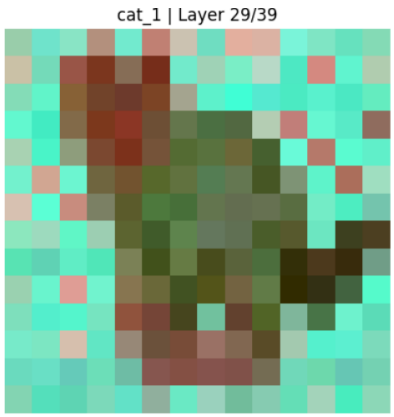
semantic abstractions (deep layers). The color shift effectively visualizes the "drift" of the image representation through the high-dimensional space. **This trajectory is visually summarized in Figure 3.8, where the subject ('Cat') transitions from a pixel-dominated red representation to a semantically abstract azure representation.**



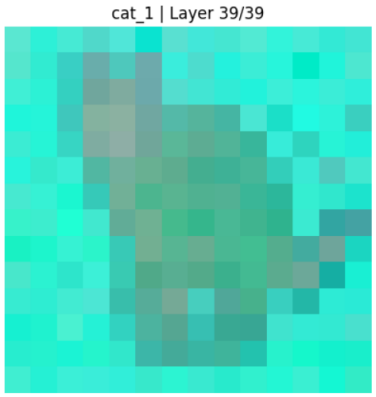
(a) : Layer 0 (Input Space)



(b) : Layer 21 (Transition)



(c) : Layer 29 (Semantic Space)



(d) : Layer 39 (Final State)

Figure 3.7: Visual evolution of the feature representation for the 'Cat' class using Global PCA (Method A). The chromatic shift from red (Layer 0) to azure (Layer 39) illustrates the global drift of feature representations described in Section 3.5.

Method A.1: Global PCA with Mean Subtraction (Centered PCA)

In standard Principal Component Analysis, it is customary to center the data by subtracting the empirical mean before calculating the covariance matrix. To investigate the effect of this centering on our feature trajectory visualization, we conducted an auxiliary experiment where the global dataset

$\mathbf{X}_{global}$ was mean-centered prior to projection:

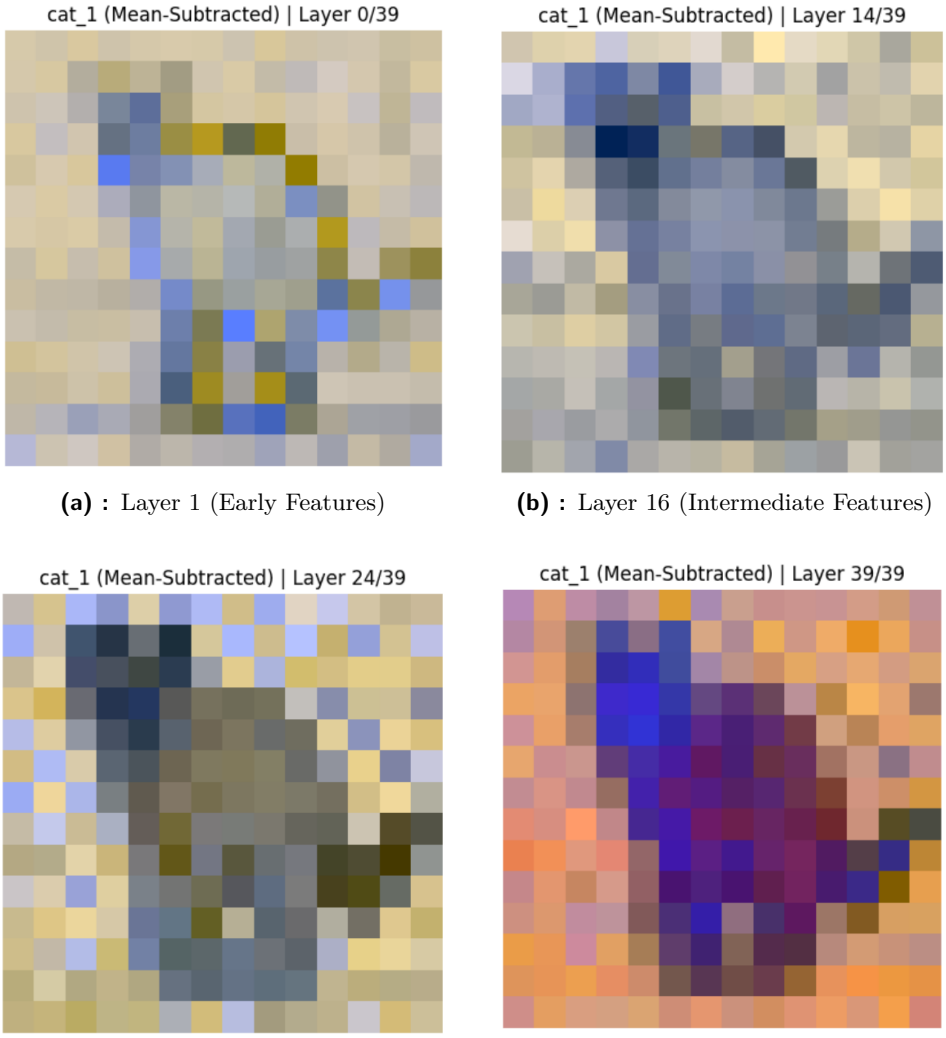
$$\bar{\mathbf{x}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i, \quad \hat{\mathbf{X}}_{global} = \mathbf{X}_{global} - \bar{\mathbf{x}} \tag{3.2}$$

where N is the total number of feature vectors across all layers l and spatial positions. The PCA estimator was subsequently fitted on the centered dataset $\hat{\mathbf{X}}_{global}$.

Visual Outcome and Analysis: Contrary to the uncentered Global PCA, applying mean subtraction resulted in a severe degradation of visual clarity. The resulting projections produced dim, desaturated, "mud-like" representations with arbitrary blue hues.

This phenomenon can be explained by the geometric behavior of feature maps in deep neural networks. As features are processed from shallow to deep layers, they exhibit a massive directional shift through the high-dimensional space. In the uncentered approach (Method A), the primary principal components naturally align with this dominant drift trajectory, mapping the large magnitudes to bright, distinct RGB values.

Conversely, when the global mean is subtracted, this primary trajectory vector collapses to the origin. The centered PCA is forced to capture only the localized variance surrounding the global centroid. Because the magnitude of this localized variance is relatively small compared to the scale of the layer-wise drift, the projected values lack the amplitude required to drive the RGB channels to bright values. The overlapping, low-magnitude variances consequently manifest as dark, ambiguous colors, with specific hues (such as blue) dominating only if a particular variance direction aligns heavily with that color channel.



(a) : Layer 1 (Early Features)

(b) : Layer 16 (Intermediate Features)

(c) : Layer 26 (Deep Features)

(d) : Layer 39 (Final Representation)

Figure 3.8: Visualization of feature maps using Global PCA with Mean Subtraction (Method A.1). Because centering collapses the global depth trajectory to the origin, the "Red-to-Azure" drift is destroyed. Instead, the PCA captures spatial variance, rendering the correlated background tokens in muddy orange/brown hues while isolating the semantic foreground (the cat) along a distinct blue principal component.

Method B: Local PCA (Contrast Maximization)

In the Local approach, we prioritize the maximum statistical separation within a specific layer, disregarding the global context. For each individual layer l, we instantiate a new PCA estimator and fit it solely on the features $\mathbf{X}^{(l)}$ of that specific layer.

$$V_{local}^{(l)} = \text{PCA}_l(\mathbf{X}^{(l)}) \tag{3.3}$$

The resulting projection is normalized independently for each frame.

Visual Outcome and Utility: While this method sacrifices the ability to track the global trajectory, it offers superior *local discriminability*.

In the Global view, the dominant variance is the **depth** signal (Input vs. Output), which can overpower subtle spatial details. By re-fitting the PCA locally, we filter out this depth signal and force the visualization to answer a specific question: *"What are the most significant differences between patches at this specific stage of processing?"*

This renders fine-grained details visible that are otherwise washed out in the Global view:

- **Shallow Layers:** It reveals high-frequency texture boundaries and edge orientations that the model is attending to.
- **Deep Layers:** It highlights subtle semantic separations (e.g., distinguishing between a dog's ear and its head) which might represent only a fraction of the total global variance but are critical for the final classification task.

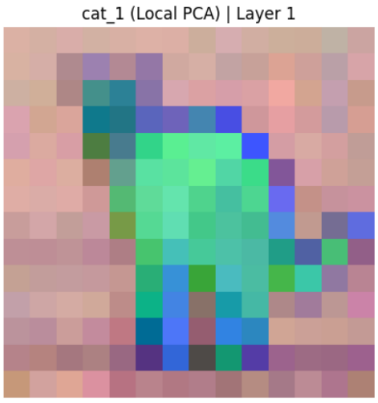
Therefore, Local PCA acts as a high-gain contrast filter, useful for inspecting the granular attention mechanisms of specific blocks without the bias of the network's overall depth trajectory.

Visual Outcome: This results in visualizations with maximum local contrast, highlighting the most salient features such as edges and textures at every specific depth. However, this method sacrifices temporal consistency. Because the eigenvectors are re-calculated at every step, the direction of the principal components can flip arbitrarily (the *sign ambiguity* problem in PCA). This manifests as abrupt, non-semantic color inversions between consecutive layers, making it difficult to track the evolution of a specific concept. **The effect of this local contrast maximization is shown in Figure 3.9. While fine-grained details are visible, the colors hold no global meaning, changing primarily due to statistical variations within the layer rather than semantic evolution.**

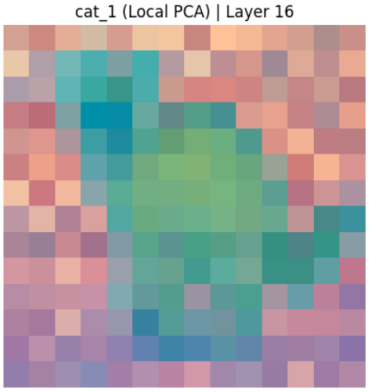
3.5.3 Comparative Analysis

The comparison between these two approaches highlights a trade-off between *local discriminability* and *global coherence*.

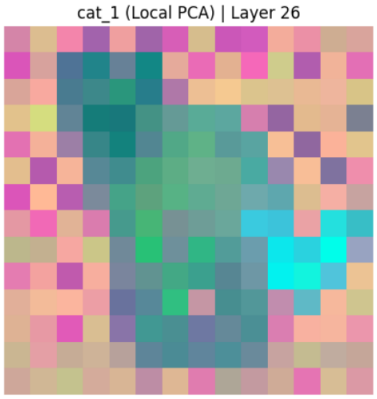
1. **Global PCA** is superior for analyzing the network's hierarchical structure. It serves as a visual proof that DinoV3 processes information sequentially, moving the data manifold from a "pixel-correlation" space to a "semantic-correlation" space [2].
2. **Local PCA** is superior for inspecting the granular attention of specific layers. It reveals that even in intermediate layers, the model separates foreground and background with high variance, although this separation is not yet aligned with the final semantic output.



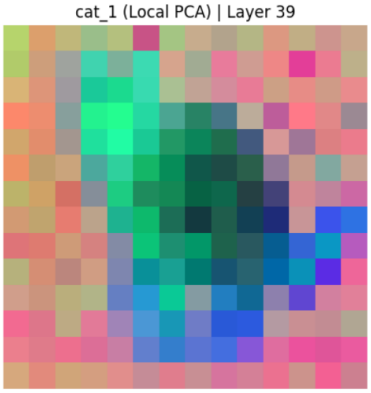
(a) : Layer 1 (Early Features)



(b) : Layer 16 (Intermediate Features)



(c) : Layer 26 (Deep Features)



(d) : Layer 39 (Final Representation)

Figure 3.9: Visualization of feature maps using Local PCA (Method B). Unlike the Global approach, dimensionality reduction is performed independently for each layer. Note that while **Intermediate Features (b)** and **Deep Features (c)** show high contrast, the loss of a shared coordinate system means the "Red-to-Azure" trajectory is no longer visible.

3.6 Spatial Attention Dynamics: The Scanning Protocol

Having analyzed the hierarchical evolution of features across the network depth (chapter 3.5). We now turn our focus to the spatial relationships encoded within the final semantic representations. Specifically, we aim to interpret the *self-attention mechanism* of the DINOv3 architecture to understand how different image regions semantically **connect** to one another.

3.6.1 Methodology: Attention Matrix Extraction

The core idea is to extract raw attention weights from the final Transformer block of our model. For a given input image, we forward pass it and intercept the self-attention weights A from the last layer. The raw attention tensor has the shape (H, N, N) where H is the number of heads and N is the sequence length (tokens).

To obtain a unified spatial map, we average the weights across all attention heads and discard the special `[CLS]` and register tokens, resulting in a patch-to-patch attention matrix $A_{patch} \in \mathbb{R}^{P \times P}$, where P is the number of spatial patches.

3.6.2 The Scanning Mechanism

The visualization is constructed as a temporal sequence where each frame represents a specific spatial query:

1. **Query Selection (The Green Box):** We define a sliding window of size $k \times k$ patches that traverses the image grid in a raster scan pattern. This window represents the region of interest Q_{pos}.

2. **Attention Integration:** For each window position, we aggregate the attention weights originating from the patches inside the window. This computes a heatmap M representing the cumulative attention of the query region:

$$M = \sum_{i \in Q_{pos}} A_{patch}[i, :] \tag{3.4}$$

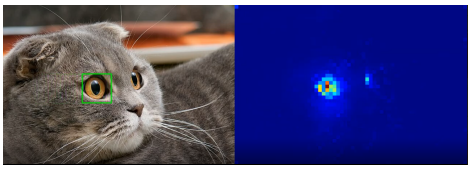
3. **Visual Overlay:** The resulting heatmap M is normalized and overlaid onto the original image.

3.6.3 Visual Interpretation

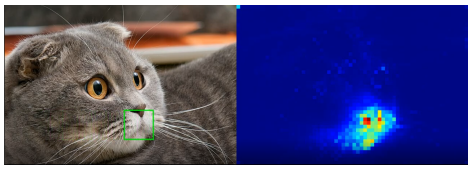
The results provide a granular view of the model's internal object segmentation. As the green query box moves over an object, the attention heatmap instantly "lights up" the entire semantic entity, demonstrating that the local patches are attending globally to the object they belong to.

Semantic similarity and Parts

In Figure 3.10, we observe the model's understanding of facial symmetry and biological features. When the query focuses on one eye, the attention map strongly activates the contralateral eye, linking them as a single semantic concept. Similarly, querying the nose region triggers attention across the whiskers, suggesting the model has learned to group these fine-grained features together.



(a) : Eye-to-Eye Correspondence



(b) : Nose-to-Whiskers Association

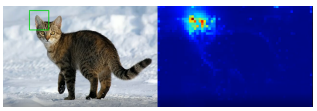
Figure 3.10: Attention dynamics on the British Shorthair. (a) Querying the left eye results in strong attention on the right eye. (b) Querying the nose spreads attention across the whisker pad.

Object Boundaries and Unity

Figure 3.11 demonstrates the model's ability to distinguish object boundaries and anatomical unity.

- **Background Awareness:** When the query is placed on the background snow (Figure 3.11a), the attention avoids the cat entirely, effectively delineating the negative space around the object boundary.
- **Anatomical Grouping:** Querying the ear (Figure 3.11b) highlights the top of the head and the opposite ear, while querying the legs (Figure 3.11c) restricts attention specifically to the limb structure.

(a) : Background Query



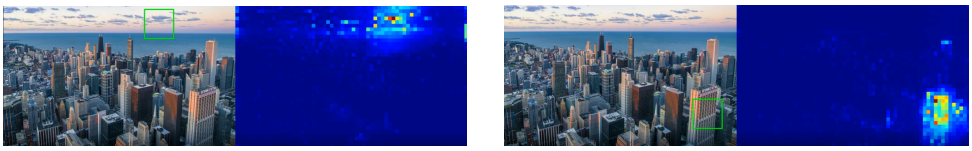
(b) : Head/Ear Query

(c) : Legs Query

Figure 3.11: (a) A background patch attends to the surroundings, respecting the object boundary. (b) An ear patch attends to the full head structure. (c) A leg patch attends specifically to the limbs.

Scene Segmentation

This behavior extends to complex scenes. In the cityscape example (Figure 3.12), querying a patch of sky activates the entire sky region while ignoring the skyline. Conversely, querying a specific skyscraper isolates that building's vertical structure from the surrounding urban clutter, demonstrating unsupervised instance-level discrimination.



(a) : Sky Segmentation **(b) :** Building Instance Isolation

Figure 3.12: (a) Querying the sky activates the atmospheric region only. (b) Querying a specific building isolates that structure from the rest of the city.

3.7 Out-of-Distribution Detection Methods

To evaluate the robustness and generalizability of the learned representations, we conducted a series of Out-of-Distribution (OOD) detection experiments. In this setup, the Food-101 dataset was designated as the In-Distribution (ID) baseline. We then evaluated the model's ability to distinguish these ID samples from three distinct OOD datasets: Oxford-IIIT Pet, FGVC Aircraft, and the Describable Textures Dataset (DTD). To preserve high-frequency visual details crucial for both fine-grained classification and texture recognition, all images were uniformly processed through a 512×512 resolution pipeline.

3.7.1 Linear Probe and Scoring Functions

Foundation models such as DINOv3 inherently output high-dimensional feature embeddings rather than class-specific logits. However, several prominent post-hoc OOD detection algorithms operate directly on the logit space of a classifier. To bridge this gap without distorting the pre-trained manifold via full fine-tuning, we trained a simple Linear Probe on top of the frozen DINOv3 embeddings. This yielded the necessary pre-softmax logits for our target ID classes (Food-101).

Using these projections, we evaluated three state-of-the-art OOD scoring mechanisms:

- **Energy Score:** Computes the Helmholtz free energy over the logit vector, functioning as a theoretically grounded alternative to the standard softmax confidence.
- **Mahalanobis Distance:** A generative, distance-based metric that measures the distance of a test sample's feature representation to the

closest class-conditional Gaussian distribution fitted on the ID training features.

- **ODIN (Out-of-DIstribution detector for Neural networks):** Utilizes temperature scaling and input preprocessing (small adversarial perturbations) to separate the softmax distributions of ID and OOD samples.

The primary evaluation metric used was the Area Under the Receiver Operating Characteristic curve (AUROC). Additionally, for qualitative analysis, we set the operating acceptance threshold to maintain a 95% True Positive Rate (TPR) for ID data.

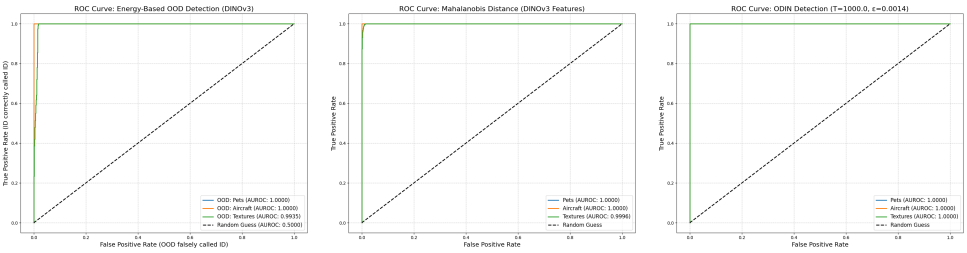
3.7.2 Results and the Semantic Overlap Anomaly

The quantitative results demonstrated near-flawless OOD detection performance (AUROC ≈ 1.0) when comparing Food-101 against the Oxford-IIIT Pet and FGVC Aircraft datasets across all three methods. The representations learned by the network cleanly separated these distinct semantic domains.

However, performance on the Textures dataset (DTD) presented a notable exception, yielding lower AUROC scores and higher False Positive Rates. Upon qualitative investigation, we discovered that this drop in statistical performance does not indicate a failure of the feature representation. Rather, it highlights a semantic overlap between the datasets: the Textures dataset inherently contains images of distinct food patterns (e.g., clustered berries, cross-sections of vegetables, woven pasta).

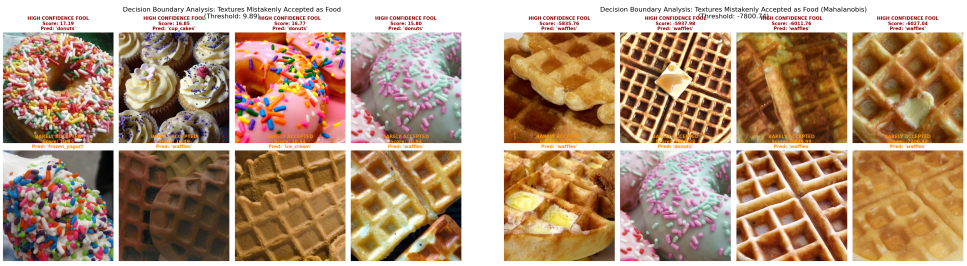
Consequently, the models confidently—and correctly, from a semantic standpoint—classified these specific OOD samples as ID (Food). Figure 3.14 illustrates the highest-confidence false positives for both the Energy Score and Mahalanobis Distance at the 95% TPR threshold. It is visually evident that the "misclassified" texture images are, in fact, food items.

Note on ODIN Evaluation: Due to the heavy computational overhead introduced by the iterative gradient-based input perturbations required for ODIN, we evaluated this specific method on a randomly sampled subset of the data. While the resulting ROC curve for ODIN (Figure 3.13) displays a perfect 100% AUROC on this specific subset, applying it to the overlapping food-texture samples yields the same semantically justified "failures" observed in the Energy and Mahalanobis methods.



(a) : Energy Score **(b) :** Mahalanobis Distance **(c) :** ODIN (Subset)

Figure 3.13: Receiver Operating Characteristic (ROC) curves for OOD detection using Food-101 as the ID dataset. The algorithms achieve flawless separation on Pets and Aircraft, while the variance in the Textures curve is driven by semantic dataset overlap.



(a) : Energy Score: Highest Confidence False Positives

(b) : Mahalanobis: Highest Confidence False Positives

Figure 3.14: Samples from the OOD Textures dataset misclassified as ID (Food-101) with the highest confidence at a 95% TPR threshold. The "errors" are technically correct visual assessments, as the images clearly depict food items acting as textures.

Chapter 4

Conclusion

This thesis explored whether the DINOv3 Vision Transformer is a suitable foundation for hierarchical image classification and Out-of-Distribution detection. We used several visualization techniques to analyze how the model understands images without being explicitly trained for these tasks.

Our experiments showed that DINOv3 creates very clear groups for natural images. In datasets like CIFAR-10 and Oxford-IIIT Pet, the model successfully keeps similar classes together. The t-SNE visualizations proved that the model is robust enough to handle different artistic styles, as seen in the Pokémon dataset. Additionally, by analyzing the network layer by layer, we confirmed that the model gradually builds understanding, moving from simple pixel patterns to meaningful concepts.

Furthermore, the attention maps revealed that the model naturally understands where objects are. It can isolate a subject from the background and group related parts without any supervision.

However, the model has clear limitations. It failed to separate classes in the SVHN dataset. This confirms that DINOv3 relies heavily on textures rather than geometric shapes. This "texture bias" is an important factor to consider when deploying the model.

In conclusion, the frozen DINOv3 model provides rich and useful features. Future work should focus on using these features to build a system that can not only detect unknown objects but also assign them to a general super-category, making AI systems safer and more interpretable.

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Appendix A

List of Abbreviations

ViT	Vision Transformer
CNN	Convolutional Neural Network
OOD	Out-of-Distribution
DINO	Self-Distillation with No Labels
ID	In-Distribution
SSL	Self-Supervised Learning
MSP	Maximum Softmax Probability
OE	Outlier Exposure
OSR	Open-Set Recognition
OWR	Open-World Recognition
ARPL	Adversarial Reciprocal Points Learning